中熔电气
SINOFUSE

# PFH-745-D-SL

## 产品规格书

## PRODUCT SPECIFICATION

SINOFUSE

成为卓越的能量保护系统解决方案服务商
为整个人类的能量转换活动的安全运行提供保障
Provide excellent integrated solution of energy protection systems
Safeguard energy conversion processes for mankind

# 文件变更履历表/Revision Record

文件名称/File Name: PFH-745-D-SL
文件编号/File No.: ZR/YC-0955

| Version | Content | Execution Date | Author |
|---|---|---|---|
| S0 | 新制定/Initial release | 2026-07-23 | Fu Juanjuan |

## 产品技术参数/Product Technical Parameters

- ●参考标准/Standards: GB 13539.7, IEC 60269-7, ISO16750/LV124
- ●额定电压/Rated voltage: DC500V
- ●额定电流/Rated current: 450A (接线铜排/Terminal busbar: 150mm²)
- ●认证/Certification: TUV
- ●符合/Compliant: RoHS
- ●阻燃等级/Flame retardant rating: UL94-V0



| NO. | Part Number | Connector Model | Connector Manufacturer | Self-triggering I²t/(A²s) | | Resistance@20±5°C (mΩ) | Weight (g) | Max Package (pcs) | Remark |
|---|---|---|---|---|---|---|---|---|---|
| | | | | Prearc | Clearing | | | | |
| 1 | PFH-745-D-SL | 3102A1004 | 健和兴/KST | 910000 | 1210000 | 0.05±0.005 | 540±10 | 12 | M8<br>12±1N·m |

注/Note:
自触发I²t的测试条件为DC300V 30kA 11μH。
The test condition for self-triggering I²t is DC300V, 30kA, 11μH.

## 产品特性/Product Features

本系列产品为无源智能熔断器，具有感知故障电流自触发分断能力，也可接受外部触发信号主动分断，具有功耗低、耐大电流冲击、ms级动作速度及全范围保护等特点，适用于道路车辆、风能、光伏及储能等行业应用，可作为电动汽车驱动系统、功率变换系统、电池、储能电容器、变频器等设备的短路、过载保护。

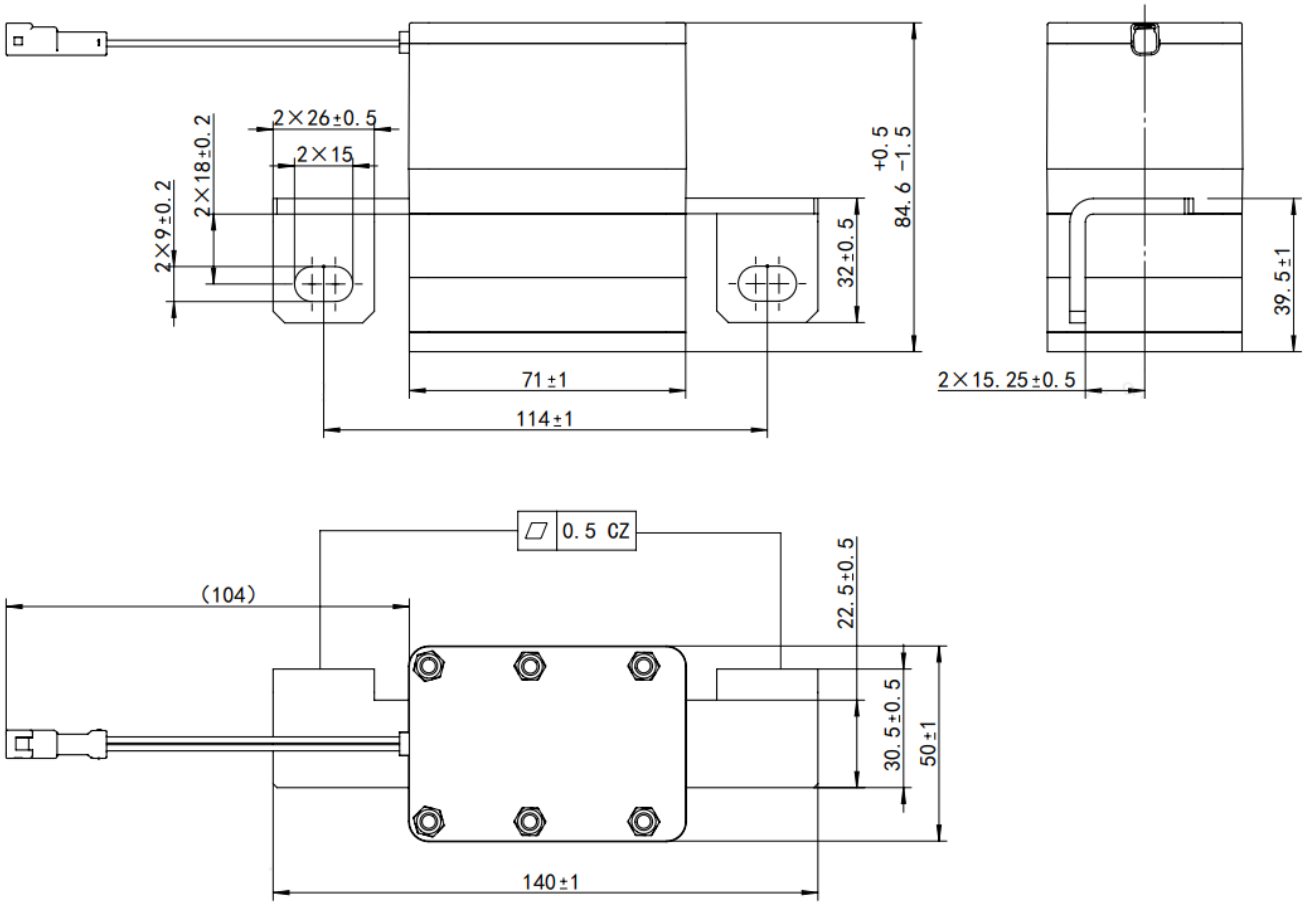
This series of products is smart fuses featuring self-triggered disconnection upon detecting fault currents, while also receiving external trigger signals for active disconnection. they offer low power consumption, high surge current withstand, millisecond-level response times, and full-range protection. suitable for applications in road vehicles, wind energy, photovoltaics, and energy storage industries, these fuses provide short-circuit and overload protection for equipment such as electric vehicle drive systems, power conversion systems, batteries, energy storage capacitors, and inverters.

## 主要技术参数/Main Technical Parameters

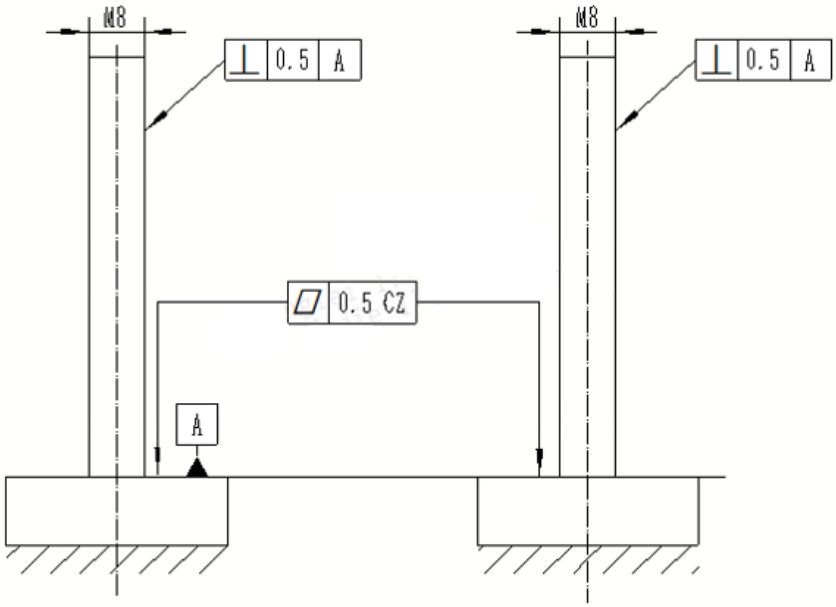
| Form | | Items | Parameters |
|---|---|---|---|
| 触发类型/Trigger Type | | 自触发&外触发/Self-triggering & external triggering | 自触发外触发一体/Integrated self-triggering and external triggering |
| 分断参数/Breaking Parameters | 自触发/Self-triggering | 最小分断能力/Minimum breaking capacity | DC500V/2kA/17μH |
| | | 最大分断能力/Maximum breaking capacity | DC300V/30kA/11μH |
| | 外触发/External triggering | 最小分断能力/Minimum breaking capacity | DC500V/0KA/17μH |
| | | 最大分断能力/Maximum breaking capacity | DC1500V/500A/电感≤700μH<br>DC1500V/500A/Inductance≤700μH |
| | | 动作时间/Action time | ≤3ms(接收到信号到切断回路/From receiving signal to cutting off circuit) |
| 绝缘电阻/Insulation Resistance | | 铜排与点火装置/Between busbar and ignition | DC1500V，分断前后/Before and after the break≥50MΩ |
| | | 铜排与外壳/Between busbar and housing | DC1500V，分断前后/Before and after the break≥50MΩ |
| | | 主铜排/Between busbar sections | DC1500V，分断后/After break≥2MΩ |
| 其它参数/Other Parameters | | 功耗/Loss | ＜20W(测试条件：环境温度20℃，导线截面积150mm²，通流450A至产品温度稳定/Test conditions: Ambient temperature 20 °C, conductor cross-sectional area 150 mm², current carrying 450 A until the product temperature stabilizes) |
| | | 耐受冲击电流/Withstand inrush current | / |

| Form | Items | Parameters |
|---|---|---|
| 触发参数/ Trigger Parameter | 点火管电阻/Igniter resistance | 1.7Ω ~ 2.3Ω |
| | 触发电流/Trigger current | ≥1.75A/0.5ms or ≥1.2A/2.0ms |
| | 不触发电流/Non-triggering current | ≤0.4A/10s |
| | 允许检测电流/Allowable detection current | ≤100mA |
| | 自触发温度/Self-triggering temperature | ≥250℃ |
| | 符合标准/Conform to standards | 满足/Qualified acc. to AK-LV16 |

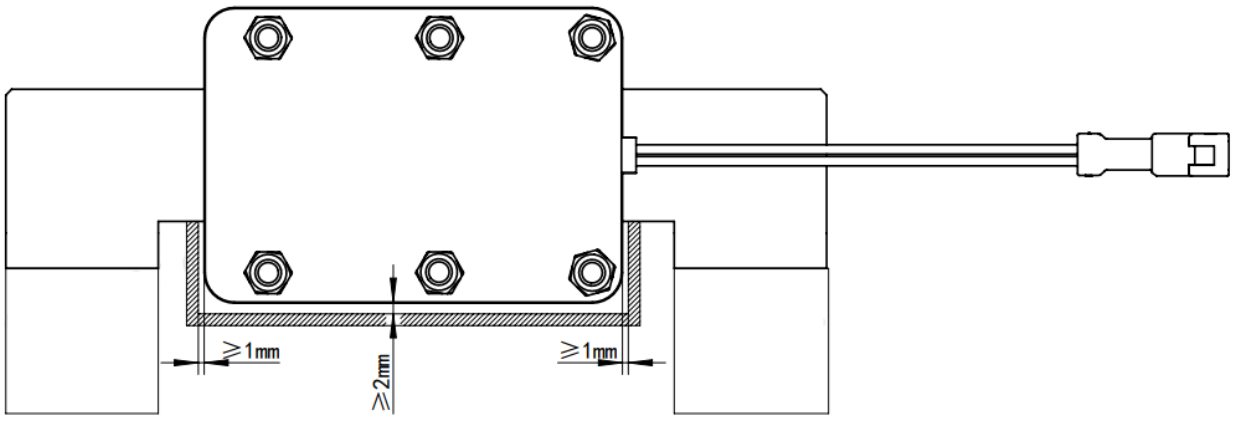
## 产品外形尺寸/Dimensions(mm)



## 推荐安装要求/Recommended Installation Requirements



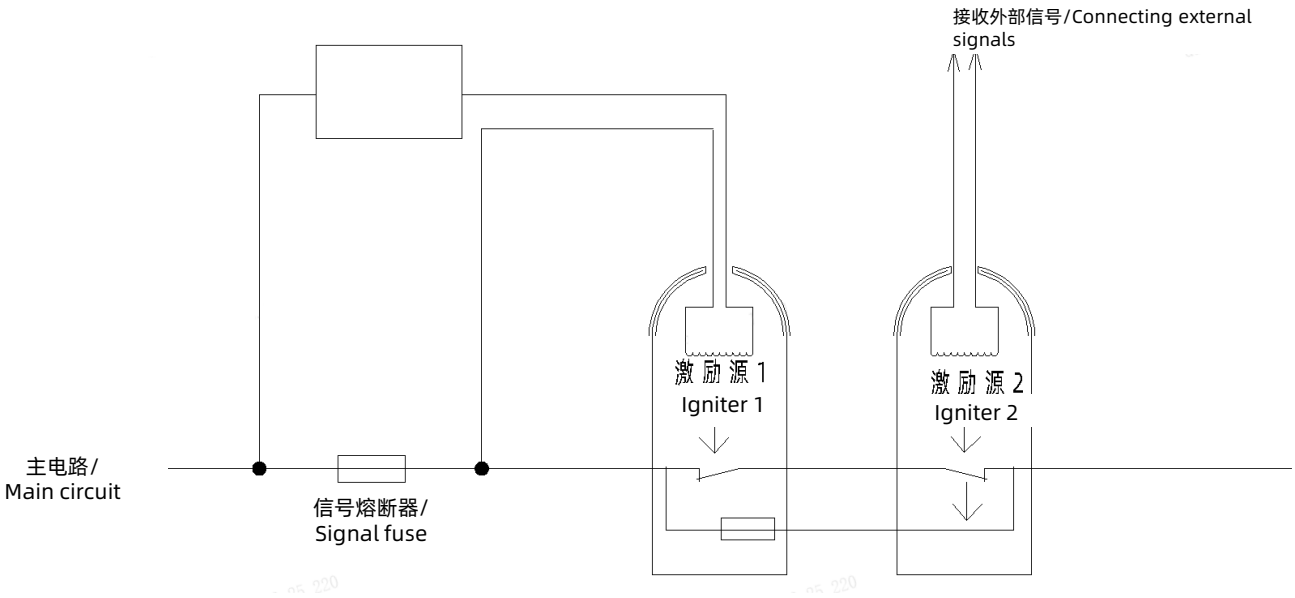

## 安装间隙要求/Installation Clearance Requirements



## 主要材质清单/Main Material List

| No. | Part Name | Material | Electroplated |
|---|---|---|---|
| 1 | 铜排/Busbar | 紫铜/Copper | GB/T 12599-Cu/Sn6b |

## 原理简介/Introduction to Principles




1.信号熔断器与激励熔断器1和激励熔断器2串联；

Signal fuse is connected in series with pyro fuse 1 and pyro fuse 2.

2.激励源1，并联在信号熔断器两端；

Igniter 1, connected in parallel across the signal fuse.

3.当信号熔断器熔断后，两端电压变大，加载在激励源1两端，触发激励源1动作，切断电路；

When the signal fuse is cut off, the voltage across it increases, This voltage is then applied across Igniter1, Igniter 1, to activate and cut off the circuit.

4.激励源2直接通过端子与外部信号连接，无中间环节，外部信号可以直接执行检测与打断；

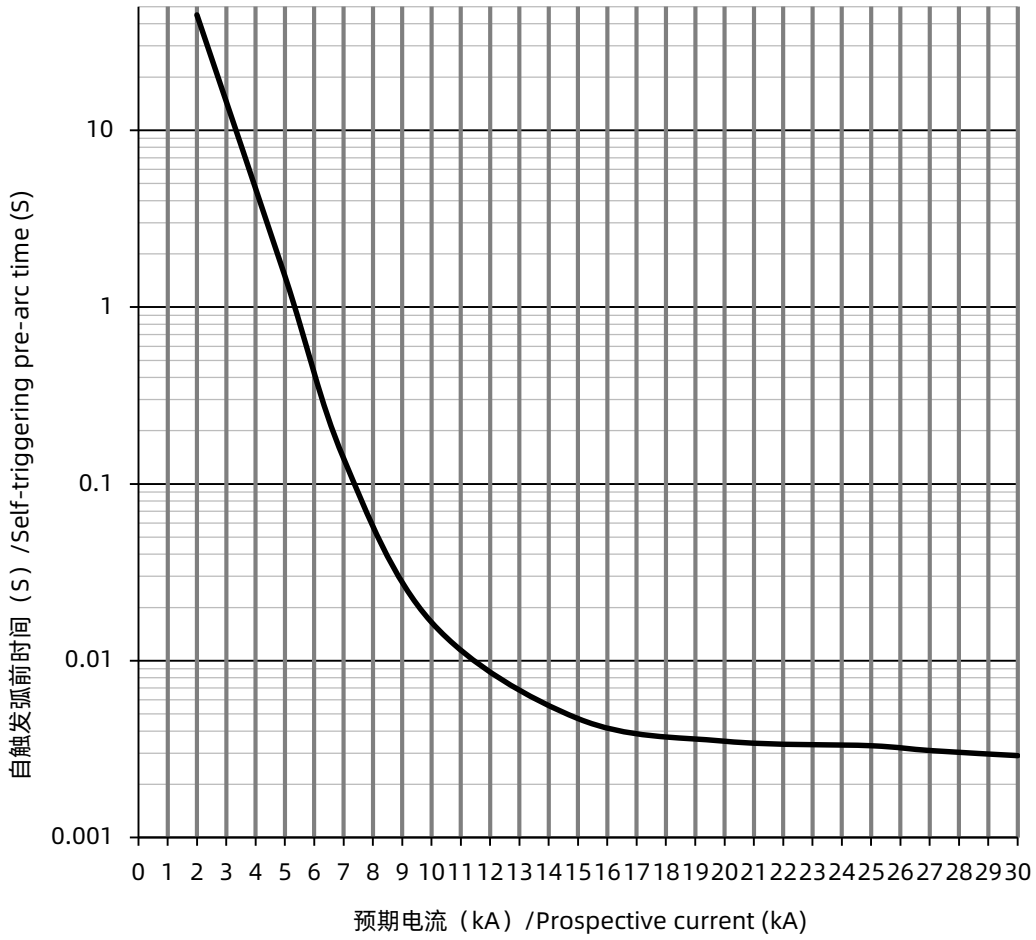
Igniter 2 connects directly to external signals via terminals, allowing external detection of the igniter status or triggering its operation to interrupt the circuit.

5.内部触发电路与外部信号完全隔离，互不干扰。

The internal trigger circuit is completely isolated from external signals, ensuring no mutual interference.

# 产品特性曲线/Characteristic Curve

**自触发动作曲线/Self-Triggering Action Curve**




注/Note:
曲线沿电流方向误差±10%。/The curve error along the current direction is ±10%.